

Leverage Staking with Liquid Staking Derivatives (LSDs): Opportunities and Risks

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Abstract

In the Proof of Stake (PoS) Ethereum ecosystem, users can stake ETH on Lido to receive **stETH**, a Liquid Staking Derivative (LSD) that represents staked ETH and accrues staking rewards. LSDs improve the liquidity of staked assets by facilitating their use in secondary markets, such as for collateralized borrowing on Aave or asset exchanges on Curve. The composability of Lido, Aave, and Curve enables an emerging strategy known as leverage staking, an iterative process that enhances financial returns while introducing potential risks. This paper establishes a formal framework for leverage staking with **stETH** and identifies 442 such positions on Ethereum over 963 days. These positions represent a total volume of 537,123 ETH (877m USD). Our data reveal that 81.7% of leverage staking positions achieved an Annual Percentage Rate (APR) higher than the APR of conventional staking on Lido. Despite the high returns, we also recognize the potential risks. For example, the Terra crash incident demonstrated that token devaluation can impact the market. Therefore, we conduct stress tests under extreme conditions of significant **stETH** devaluation to evaluate the associated risks. Our simulations reveal that leverage staking amplifies the risk of cascading liquidations by triggering intensified selling pressure through liquidation and deleveraging processes. Furthermore, this dynamic not only accelerates the decline of **stETH** prices but also propagates a contagion effect, endangering the stability of both leveraged and ordinary positions.

1 Introduction

The Ethereum blockchain’s transition from Proof-of-Work (PoW) [1] to Proof-of-Stake (PoS) [2, 3, 4, 5] is a remarkable shift towards a more sustainable consensus mechanism. This change, while crucial for energy efficiency, introduces new challenges in staking ETH. In PoS-based Ethereum, validators must stake ETH to secure the network [6, 7] and earn staking rewards. However, solo staking demands a substantial capital commitment of 32 ETH and technical expertise in maintaining a validator node. Additionally, staked ETH becomes illiquid during the staking period, limiting its usability for other financial activities.

To mitigate these challenges, Liquid Staking Derivatives (LSDs), also referred to as Liquid Staking Tokens (LSTs), have emerged as transformative solutions. These assets enhance the liquidity of staked assets while preserving their earning potential. Retail users can flexibly stake any amount of ETH on a liquid staking platform (e.g., Lido) to receive the corresponding LSDs. These LSDs are fungible and tradable representations of the staked ETH and its associated rewards. At the time of writing, Lido stands as a leading LSD provider on Ethereum, marked by its top position with a Total Value Locked (TVL) of 38b USD¹.

¹<https://defillama.com/protocol/lido>, last accessed on Dec 5, 2024.

In the LSD primary market, platforms such as Lido allow users to convert ETH into **stETH** for various uses within the Decentralized Finance (DeFi) ecosystem. Specifically, users might choose to simply hold **stETH** to accrue a staking Annual Percentage Rate (APR) of around 3.6% (as of Mar 2024) or utilize **stETH** in the secondary market for further financial activities. For example, **stETH** can serve as collateral on DeFi lending platforms such as Aave to borrow ETH. This allows users to earn rewards on their staked ETH while utilizing **stETH** as active investment capital². Additionally, **stETH** can be used for liquidity provision and traded for ETH in the **stETH**–ETH pool of a Decentralized Exchange (DEX), such as the Curve protocol.

The composability between Lido, Aave, and Curve facilitates two novel strategies of leverage staking (see Figure 2 and 3). The first, known as *direct leverage staking*, involves users staking ETH on Lido in the primary market to receive **stETH**, which is then used as collateral on Aave to borrow ETH, subsequently restaked on Lido. Users can iteratively execute this process to increase financial returns based on their risk profile. The second strategy, *indirect leverage staking*, involves initially swapping ETH for **stETH** within the Curve pool at secondary market prices, then using the acquired **stETH** as collateral on Aave to borrow more ETH, which is swapped again for **stETH** in the Curve pool. This allows users to participate in staking and earn rewards without directly staking their ETH on Lido. Together, these strategies demonstrate the flexibility and depth of the LSD ecosystem, offering varied approaches to increasing returns with leveraged positions.

While leverage staking offers high return opportunities, it also presents potential risks. Under adverse market conditions that lead to a substantial decline in **stETH** prices, leverage staking can act as a catalyst for market instability by increasing the risk of *cascading liquidations*, where successive liquidations drive **stETH** prices into a downward spiral. As such, this paper aims to understand the opportunities and risks of leverage staking. We investigate the underlying mechanisms, evaluate the financial benefits and inherent risks of this strategy, and assess its broader market impact. We outline our main contributions as follows:

Strategy Formalization. We develop a formal framework for leverage staking with **stETH** that captures direct and indirect strategies. We conduct an analytical study to derive key metrics such as leverage staking multiplier, Health Factor (HF), and APR for each position. To the best of our knowledge, we are the first to model leverage staking with LSDs.

Empirical Measurement. We empirically analyze leverage staking spanning 963 days, from Dec 17, 2020 to Aug 7, 2023. We detect 262 direct leverage staking positions with a total stake amount of 295,243 ETH (482m USD), and 180 indirect leverage staking positions with a total swap amount of 241,880 ETH (395m USD). We observe that a majority (81.7%) of leverage staking positions yielded an APR higher than conventional staking on Lido.

User Behavior Analysis. We explore the **stETH** price deviation in the context of the Terra crash incident. We analyze how users behave when faced with potential liquidations. We discover that users actively deleveraged their leverage staking positions and collectively repaid a substantial debt of 136,069 ETH, further intensifying the **stETH** selling pressure.

Stress Testing. We perform stress tests on the Lido–Aave–Curve LSD ecosystem to evaluate the impact of leverage staking under extreme conditions of significant **stETH** devaluation. Our simulation reveals that leverage staking escalates the risk of cascading liquidations. Systemic risk is exacerbated not only through liquidations but also via the market pressures these actions generate. The selling pressure on **stETH**, driven by both liquidations and deleveraging actions, can trigger a contagion effect across the system, further depressing **stETH** prices and adversely impacting the stability of broader market participants, including ordinary users.

²<https://github.com/lidofinance/aave-asteth-deployment>

2 Related Work

We provide an overview of the literature on PoS staking, LSDs, and DeFi lending.

PoS Staking. The economics of PoS staking has been explored in prior research works. For example, Cong *et al.* [8] developed a continuous-time model to explore the economic impact of staking in token-based digital economies. They found that higher staking rewards lead to increased staking ratios, which in turn predict higher token price appreciation and generate profitable carry trade opportunities with significant Sharpe ratios. Additionally, attention has been given to the security of PoS staking. For instance, Chitra [9] investigated how on-chain lending affects the security of a PoS blockchain. They found that when the yield from lending contracts is higher than the inflation rate from staking, stakers are incentivized to remove their staked tokens and lend them out, thus reducing network security.

LSDs. Tzinis *et al.* [10] studied the Principal-Agent problem in the liquid staking setting. They discussed the dilemma between the choice of proportional representation and fair punishment and proposed a concrete attack to illustrate their incompatibility. Scharnowski *et al.* [11] analyzed the liquid staking basis (e.g., the discrepancy) between the prices of LSDs in the primary and secondary market. They observed that the liquid staking basis widens when cryptocurrency volatility increases and liquidity decreases in the secondary market. Cintra *et al.* [12] utilized the Bayesian Online Changepoint Detection algorithm to identify potential depeg incidents using price data from the Curve `stETH-ETH` pool. They show that the proposed approach can help users manage potential risks.

DeFi Lending. Heimbach *et al.* [13] studied the impact of the Ethereum merge on two DeFi lending platforms, Compound and Aave. They investigated the actions taken by Aave to mitigate the liquidation risk of collateralized `stETH` positions. Wang *et al.* [14] formalized a model for under-collateralized DeFi lending platforms and empirically evaluated the risks associated with leveraging, such as impermanent loss, arbitrage and liquidation.

3 Background

3.1 Blockchain and DeFi

Blockchain is a decentralized digital ledger that records transactions across multiple nodes to ensure security, transparency, and immutability. It is the foundational technology behind cryptocurrencies such as Bitcoin [15] and Ethereum [1], enabling peer-to-peer (P2P) transactions without the need for a trusted third party. The blockchain structure as a series of blocks chained together through cryptographic hashes helps prevent alteration to the data once it has been confirmed on-chain. A permissionless blockchain allows any participant to join and engage without requiring authorization. In this context, the Ethereum blockchain [16] emerges as a pioneering platform, supporting the execution of smart contracts and empowering developers to create various decentralized applications.

DeFi [17] represents an innovative application of blockchain technology, focusing on building open financial systems. DeFi refers to a set of blockchain-based financial services and products that operate without intermediaries, using smart contracts to build an open environment. DeFi innovations ranging from collateralized lending to DEXs are reshaping the financial system. The TVL in DeFi hit a record high of 178b USD in Nov 2021, with the Ethereum blockchain driving these DeFi activities.

3.2 Ethereum PoS

The PoS consensus mechanism, first proposed in online forums and later examined by academia [18, 6, 19, 20, 21], has emerged as an energy-efficient alternative to PoW.

Beacon Chain. On Dec 1, 2020, Ethereum marked a significant milestone by introducing its PoS-based Beacon Chain that runs in parallel with Ethereum’s PoW Mainnet. In the Beacon Chain, “staking” is introduced through a deposit mechanism, allowing participants to become validators by locking up 32 ETH in a designated smart contract. Staking enables validators to contribute to the integrity of the network by participating in the consensus process, including proposing and validating blocks. This not only helps maintain the security of the blockchain, but also allows stakeholders to earn rewards proportional to their contributions, incentivizing more participants to engage in the network’s operation.

The Merge. On Sep 15, 2022, the Merge enables Beacon Chain to evolve as the consensus mechanism for the entire Ethereum network [22]. Ethereum now runs on the execution layer and the consensus layer. The execution layer is responsible for executing transactions, defining how the state of the Ethereum network changes over time. The role of the consensus layer entails establishing agreement among validators regarding the state of the execution layer. The Ethereum staking system offers various incentives to validators. Rewards from the consensus layer include block proposal, attestation, and sync committee rewards [2]. The execution layer introduces additional rewards, including priority tips and Maximal Extractable Value tips [23, 24]. Penalties also apply to dishonest behaviors.

The Shapella Upgrade. On Apr 12, 2023, Ethereum underwent the “Shapella upgrade”. The Shapella upgrade combines the “Shanghai upgrade” and the “Capella upgrade”, which took place on the consensus and execution layer simultaneously [25]. The Shapella upgrade primarily introduces the capability to unstake ETH secured within the network. This ability enhances the operational dynamics for both individual stakers and validators. Validators can initiate withdrawals of their staked ETH, either partially or in full, enabling them to reclaim their capital and potentially redeploy it elsewhere. Similarly, if a user has staked ETH on Lido (see Section 3.4), they now have the flexibility to partially or fully unstake their assets, allowing for greater liquidity and control over their investments.

3.3 Staking Options

Ethereum participants are presented with four distinct staking options as follows.

Solo Staking. In solo staking, individual participants operate their validator nodes by committing a threshold of 32 ETH, thus maintaining full control over the staking rewards. However, solo staking necessitates technical expertise to manage a validator node. Furthermore, its substantial capital requirement may render it financially inaccessible for many retail users.

Staking as a Service (SaaS). For users possessing the requisite 32 ETH but lacking in technical expertise, SaaS presents a viable solution. SaaS manages the validator node on behalf of users, utilizing their signing keys to perform on-chain tasks³. This simplifies the staking process for individuals and mitigates the risks associated with node management.

Pooled Staking. For retail users with holdings below the 32 ETH threshold, pooled staking emerges as a feasible alternative, enabling them to collectively participate in the network’s validation process, earn rewards, and capitalize on the broader Ethereum ecosystem without the need for individual, full-node commitments. Typically, staking pools charge fees, which

³<https://ethereum.org/en/staking/saas/>

are further split between Node Operators (NOs) and the protocol Decentralized Autonomous Organization (DAO). NOs run and maintain validator nodes on behalf of the staking pool, while the DAO selects NOs and configures crucial parameters for the protocol.

Centralized Exchange (CEX) Staking. CEXs, such as Coinbase and Binance, provide centralized and custodial staking services to users. These services simplify the staking process by managing technical aspects and providing a user-friendly interface. However, such convenience comes with inherent risks associated with the centralized nature of CEX staking. Users must trust these platforms with their assets, making them vulnerable to security breaches, regulatory changes, or operational failures.

3.4 LSD

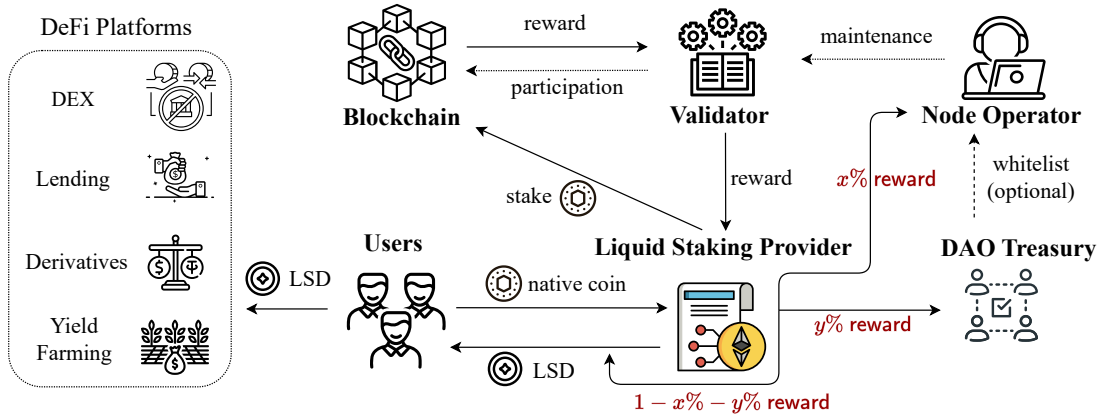


Figure 1: Overview of the LSD Ecosystem.

Staking offers several advantages, from earning rewards to enhancing network security. However, once ETH is locked for staking, it becomes illiquid, making it inaccessible for trading. Given this challenge, the concept of LSD emerged, which represents staked assets and rewards in a tradable form. Figure 1 provides an overview of the LSD ecosystem. When users stake ETH within an LSD provider (e.g., a liquid staking pool), they receive LSDs in return.

At the time of writing, liquid staking protocols accumulate a TVL of more than 65b USD⁴, securing the top position in TVL across various DeFi sectors. Users can obtain LSDs through two primary staking methods: pooled staking and CEX staking. Pooled staking protocols such as Lido, Rocket Pool, Frax, Stakewise, and Swell Network provide LSDs to users. CEXs such as Coinbase and Binance also support LSDs.

Lido is currently the leading LSD provider and ranks as the largest DeFi protocol in terms of TVL. Lido is recognized as the leading LSD provider and ranks as the largest DeFi protocol in terms of TVL. The total amount of ETH staked on Lido reached 9.67m in Dec 2024, accounting for 70.5% of the total ETH LSDs on Ethereum. Users stake ETH on Lido to receive **stETH** in return. **stETH** implements the *rebasing mechanism*, where **stETH** holders' account balances get adjusted daily to reflect the accumulated rewards [26]. The rebase can be positive or negative, depending on the validators' performance.

⁴<https://defillama.com/protocol/lido>, last accessed on Dec 5, 2024

3.5 DeFi Lending Protocols

DeFi lending protocols are decentralized platforms that facilitate P2P lending and borrowing of cryptocurrency assets through the automated execution of smart contracts. At the time of writing, Aave stands as the leading DeFi lending protocol, with a total TVL of 36b USD⁵. The Aave V2 lending protocol follows an over-collateralization model, meaning that users must supply more collateral value than the borrowed amount. As an illustration, when the collateral value amounts to S ETH, the user’s borrowing capacity is restricted to no more than $S \cdot l$ ETH, where $l \in [0, 1]$ denotes the Loan-to-Value (LTV) ratio. In the event that the collateral value falls below a specified threshold, users may need to add more collateral or risk the liquidation of their asset to repay the borrowed amount and accrued interest. To monitor the collateralization status of each position, Aave utilizes Liquidation Threshold (LT) to establish the threshold percentage that designates a position as under-collateralized, and HF as a key metric to quantify the liquidation status of a position. Specifically, user U_i ’s position can be liquidated if $HF_{(U_i)} < 1$ (see Equation 1).

$$HF_{(U_i)} = \frac{\sum_j \text{collateralized value of asset}_j \text{ in ETH} \cdot LT_j}{\sum_j \text{borrowed value of asset}_j \text{ in ETH}} \quad (1)$$

4 System Model

We first provide a set of notations to facilitate the understanding of related equations.

Notation	Meaning	Notation	Meaning
S	Initial investment (principal amount) in ETH	$p_{t_0}^{1st}$	stETH price in the primary market at t_0
l	The Loan-to-Value (LTV) ratio used by Aave	$p_{t_0}^{2nd}$	stETH price in the secondary market at t_0
LT	The liquidation threshold used by Aave	$p_{t_0}^a$	stETH price used by Aave at time t_0
HF_{U_i}	The health factor (HF) of U_i ’s position	$p_{t_c}^a$	stETH price used by Aave at time t_c

Table 1: Notations used to formalize the leverage staking strategy.

4.1 System Participants

We consider an LSD ecosystem on the Ethereum blockchain with the following participants.

Users: A user (U_i) is depicted as a rational and strategic entity, proficient in interacting with multiple DeFi platforms. U_i can adopt diverse strategies to maximize financial returns.

Liquid Staking Providers: U_i can stake ETH on liquid staking platforms (e.g., Lido) to receive LSDs (e.g., stETH). These LSDs can then be used in various financial activities, including trading, collateralized borrowing, and liquidity provision.

Lending and Borrowing Providers: U_i can supply a single asset on DeFi lending platforms such as Aave, using it as collateral to secure a loan in the form of a different asset.

4.2 Leverage Staking with LSDs

This section introduces and compares three strategies: (i) leverage borrowing, (ii) direct leverage staking, and (iii) indirect leverage staking strategies.

⁵<https://defillama.com/protocol/aave>, last accessed on Dec 5, 2024.

Investment Strategy	Asset Pair	Leverage Pattern	Staking Reward	Deposit-Borrow Revenue
Leverage Staking (Direct)	ETH-LSD	Stake→Deposit→Borrow→Stake	✓	✓
Leverage Staking (Indirect)	ETH-LSD	Swap→Deposit→Borrow→Swap	✓	✓
Leverage Borrowing	Non-LSD Pair	Swap→Deposit→Borrow→Swap	✗	✓

Table 2: Comparison of leverage staking and leverage borrowing strategies on Ethereum.

4.2.1 Leverage Borrowing

In Ethereum, *leverage borrowing* emerges as a commonly adopted strategy. This process involves users initially exchanging asset X for Y within a DEX pool. Subsequently, asset Y is supplied as collateral on lending platforms to borrow asset X. This borrowed asset X is then exchanged again for Y in the DEX pool, enabling users to iteratively amplify their leverage borrowing positions. The financial incentive behind the leverage borrowing strategy lies in the user’s ability to expand the deposit-borrow leverage through repeated swap, deposit, and borrow actions. Although the deposit rate offered by the lending platform is lower than the borrowing rate, the larger total amount of asset Y deposited compared to asset X borrowed typically results in net earnings, which can be further amplified through leverage.

4.2.2 Leverage Staking

Following the introduction of PoS staking, a concept analogous to leverage borrowing, termed *leverage staking*, has emerged. It is a strategy linked with LSDs and involves a recursive cycle of stake/swap, deposit, and borrow actions (see Figure 2 and 3) to increase financial returns. We concentrate our analysis on leverage staking within the Lido–Aave–Curve ecosystem and describe direct and indirect leverage staking as follows.

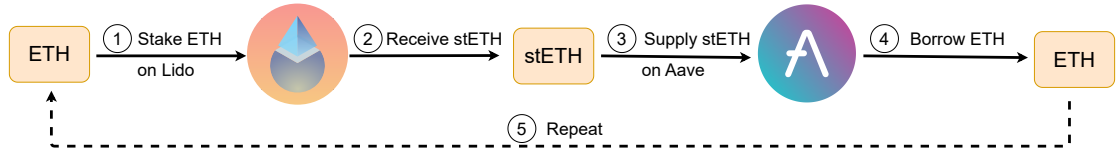


Figure 2: Overview of the direct leverage staking strategy.

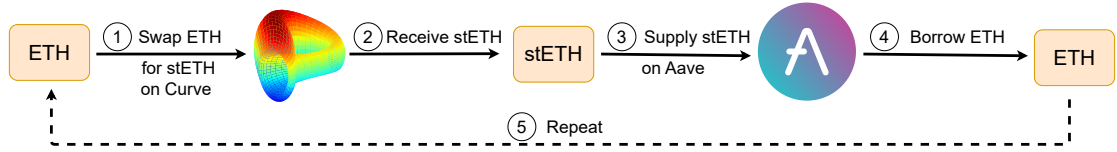


Figure 3: Overview of the indirect leverage staking strategy.

Direct Leverage Staking. U_i first stakes a principal amount of S ETH on Lido at time t_0 to acquire $S/p_{t_0}^{1st} = S$ stETH, where $p_{t_0}^{1st} = 1$ denotes the stETH to ETH price in the primary market. Next, U_i supplies stETH on Aave to borrow $S \cdot l \cdot p_{t_0}^a$ amount of ETH, where l denotes the LTV ratio and $p_{t_0}^a$ denotes the stETH to ETH price used by Aave V2 lending protocol⁶.

⁶The Aave V2 lending protocol uses the Chainlink price oracle. See <https://etherscan.io/address/0xa50ba011c48153de246e5192c8f9258a2ba79ca9#code>.

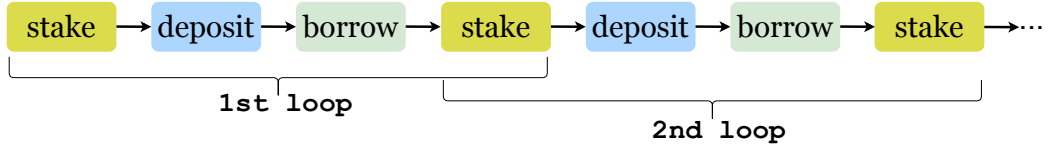


Figure 4: The illustration of direct leverage staking loops. The user completes the k^{th} loop via a sequence of actions: $\{\text{stake}, \text{deposit}, \text{borrow}, (\text{re})\text{stake}\}$. In parallel, an indirect leverage stake loop is characterized by the sequence $\{\text{swap}, \text{deposit}, \text{borrow}, (\text{re})\text{swap}\}$. Within these frameworks, the $(\text{re})\text{stake}/(\text{re})\text{swap}$ is crucial in completing the respective loops.

Then U_i restakes the borrowed ETH on Lido. U_i performs this *loop* for n times, amplifying both the staking reward from Lido and the deposit-borrow revenue from Aave.

Indirect Leverage Staking. Instead of acquiring stETH from Lido, U_i can first swap S ETH for $S/p_{t_0}^{\text{2nd}}$ stETH ⁷ within the Curve pool at time t_0 , where $p_{t_0}^{\text{2nd}}$ denotes the stETH to ETH price in the secondary market. Subsequently, U_i supplies stETH on Aave to borrow $S \cdot p_{t_0}^a \cdot l / p_{t_0}^{\text{2nd}}$ ETH. The borrowed ETH is swapped again for stETH within the Curve pool. Although not engaging in direct staking on Lido, U_i can still accrue staking rewards, as stETH employs a rebasing mechanism for reward distribution (see Section 3.4).

Leverage Multiplier. Assume U_i invests a total asset (staked ETH on Lido or swapped ETH within Curve) of $A_{(S,n)}$ ETH through leverage staking with an initial investment of S ETH, the leverage multiplier $LevM_{(S,n)} = \frac{A_{(S,n)}}{S}$ is defined as the ratio between $A_{(S,n)}$ and S .

Direct vs Indirect Leverage Staking. Both direct and indirect leverage staking strategies are designed to amplify staking rewards through a recursive methodology. However, direct leverage staking acquires stETH from the LSD primary market, whereas indirect leverage staking obtains stETH from the secondary market. Notably, indirect leverage staking bypasses the staking process, consequently not contributing to the increase in the total staked ETH within the network. Utilizing the same principal amount of ETH, U_i generally acquires more stETH through the indirect strategy, as the stETH to ETH price in the secondary market is often below 1. However, this approach incurs additional costs such as price slippage.

Leverage Staking vs Leverage Borrowing. While leverage staking and leverage borrowing both exhibit recursive patterns, they differ for several reasons. First, leverage staking primarily targets LSDs, whereas leverage borrowing is focused on non-LSD tokens. Second, leverage staking aims to amplify both staking rewards and the deposit-borrow revenue, while leverage borrowing solely seeks to enhance the deposit-borrow revenue (see Table 2). Third, they bear different risk sources. In addition to market risk, leverage staking involves the risk of slashing (see Section 3.2). To summarize, although the traditional leverage borrowing strategy shares some similarities with the leverage staking strategy discussed in this paper, they differ in terms of underlying assets, income sources, and associated risk.

5 Analytical Study

This section conducts an analytical study on the leverage staking strategy. We also offer a generalized formalization encompassing other potential scenarios in Appendix B.

We assume that U_i can complete n loops (see Figure 4) within a short time interval such that the stETH price remains unchanged. As a rational participant, U_i determines the value

⁷Ignore swap fees for illustration purposes.

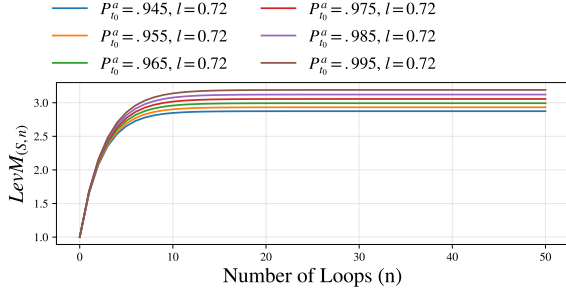


Figure 5: $LevM_{(S,n)}$ with varying $p_{t_0}^a$.

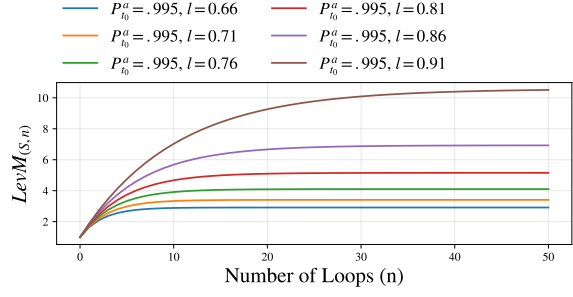


Figure 6: $LevM_{(S,n)}$ with varying l .

of n according to its risk profile. Let $p_{t_0}^a$ denote the Aave lending price of **stETH** and $p_{t_0}^m$ be the **stETH** to **ETH** market price, where $p_{t_0}^m = p_{t_0}^{1st}$ for direct leverage staking and $p_{t_0}^m = p_{t_0}^{2nd}$ for indirect leverage staking. U_i acquires a total investment amount of $A_{(S,n)}$ **ETH**, collateral amount of $C_{(S,n)}$ **stETH** and debt amount of $B_{(S,n)}$ **ETH** (see Equation 2).

$$\begin{aligned}
 A_{(S,n)} &= S \cdot \left[1 + \frac{l \cdot p_{t_0}^a}{p_{t_0}^m} + \dots + \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^n \right] = S \cdot \frac{1 - \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^{n+1}}{1 - \frac{l \cdot p_{t_0}^a}{p_{t_0}^m}} \\
 C_{(S,n)} &= \frac{S}{p_{t_0}^m} \cdot \left[1 + \frac{l \cdot p_{t_0}^a}{p_{t_0}^m} + \dots + \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^{n-1} \right] = \frac{S}{p_{t_0}^m} \cdot \frac{1 - \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^n}{1 - \frac{l \cdot p_{t_0}^a}{p_{t_0}^m}} \\
 B_{(S,n)} &= S \cdot \left[\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} + \dots + \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^n \right] = S \cdot \frac{\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} - \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^{n+1}}{1 - \frac{l \cdot p_{t_0}^a}{p_{t_0}^m}}
 \end{aligned} \tag{2}$$

Leverage Multiplier. In Equation 3, the leverage multiplier is defined as the ratio of $A_{(S,n)}$ to S . For direct leverage staking, we use the primary market price ($p_{t_0}^m = p_{t_0}^{1st} = 1$). Consequently, $LevM_{(S,n)}$ simplifies to $(1 - (l \cdot p_{t_0}^a)^{n+1}) / (1 - l \cdot p_{t_0}^a)$. Figure 5 and 6 show how $LevM_{(S,n)}$ changes in response to variations in the **stETH** price use by Aave ($p_{t_0}^a$) and the LTV ratio (l) respectively. Notably, when looping towards infinity, the value of $LevM_{(S,n)}$ converges to $1 / (1 - l \cdot p_{t_0}^a)$. For indirect leverage staking where $p_{t_0}^m = p_{t_0}^{2nd}$, when $\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} < 1$, $LevM_{(S,n)}$ converges towards $1 / (1 - \frac{l \cdot p_{t_0}^a}{p_{t_0}^m})$ as the number of loops approaches infinity.

$$LevM_{(S,n)} = \frac{A_{(S,n)}}{S} = \frac{1 - \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^{n+1}}{1 - \frac{l \cdot p_{t_0}^a}{p_{t_0}^m}} \tag{3}$$

Health Factor. Leverage staking can yield high returns but also raises the risk of liquidation. U_i 's position may be susceptible to liquidation if the value of the collateralized **stETH** declines. As discussed in Section 3.5, Aave uses HF to track the status of each position. U_i 's position can be liquidated if $HF < 1$ (see Equation 4). In our leverage staking example, the parameters l and LT are 69% and 81% respectively (see Table 3 for the changes of Aave parameter configurations). Historically, the Aave v2 always ensures $l < LT$.

Let $\Delta\%p_{\Delta t}^a$ denotes the percentage change of Aave **stETH** price from t_0 to t_c . $\Delta\%p_{\Delta t}^a$ is negative if **stETH** price decreases. When **stETH** price increases, Equation 5 always holds.

When **stETH** price decreases, Equation 5 suggests that, to uphold a secure position with $HF > 1$, the largest acceptable percentage decrease in **stETH** price is $\frac{l}{LT} - 1 = -\frac{12}{81} \approx -14.8\%$. In a liquidation event, the user’s entire collateralized **stETH** will be liquidated, and this effect becomes more pronounced as the number of loops (n) increases.

$$HF_{U_i}(p_{t_c}^a | p_{t_0}^a) = \frac{\sum_n^k k^{th} \text{ collateralized stETH value in ETH} \cdot LT}{\sum_n^k k^{th} \text{ borrowed ETH value}} = \frac{C(S, n) \cdot p_{t_c}^a \cdot LT}{B(S, n)} = \frac{p_{t_c}^a \cdot LT}{p_{t_0}^a \cdot l} \quad (4)$$

$$HF_{U_i}(p_{t_c}^a | p_{t_0}^a) \geq 1 \implies \frac{p_{t_c}^a}{p_{t_0}^a} > \frac{l}{LT} \implies \Delta\%p_{\Delta t}^a \geq \frac{l}{LT} - 1 \quad (5)$$

Profit Breakdown. Equation 6 calculates leverage staking profitability. Let r_s , r_c , r_b represent the staking APR offered by Lido and the deposit and borrow interest rates provided by Aave respectively. It is worth noting that r_s changes in accordance with the validator performance, while r_c and r_b vary based on Aave’s interest rate model⁸. U_i earns a staking APR of $R_s(n)$ and a deposit APR of $R_c(n)$, while pays a borrow APR of $R_b(n)$. In the case of leverage staking, the factor by which $R_s(n)$ is amplified is the total amount of the investment divided by the initial amount of the investment (i.e., $\frac{A_{(S,n)}}{S} = LevM_{(S,n)}$). Similarly, the factor by which $R_c(n)$ is amplified is the total amount of the collateral divided by the initial amount of the collateral (i.e., $\frac{C_{(S,n)}}{S/p_{t_0}^m}$). The same logic applies to $R_b(n)$. As such, U_i obtains a net APR of $R_{Net}(n) = R_s(n) + R_c(n) - R_b(n)$. The necessary condition for a rational U_i to apply leverage staking rather than conventional staking is $R_{Net}(n) > r_s$.

$$R_{Net}(n) = R_s(n) + R_c(n) - R_b(n) = r_s \cdot \frac{A_{(S,n)}}{S} + r_c \cdot \frac{C_{(S,n)} \cdot p_{t_0}^m}{S} - r_b \cdot \frac{B_{(S,n)} \cdot p_{t_0}^m}{S \cdot l \cdot p_{t_0}^a} \quad (6)$$

In addition to the standardized scenario discussed above, real-world applications of leverage staking can vary significantly among users. For instance, a user might choose not to reinvest all of their received **stETH** on Aave. For a more detailed exploration of this variability, please see the generalized formalization in Appendix B.

6 Empirical Study

We outline the empirical evaluation of leverage staking across Aave, Lido and Curve.

Data Collection. We first crawl the on-chain events involving users’ actions on the Aave V2 lending pool, including **deposit**, **borrow**, **withdraw**, and **repay** events. For direct leverage staking, we crawl the historical **stake** (i.e., **submitted**) events related to Lido **stETH** token when users stake ETH on Lido. For indirect leverage staking, we crawl the historical **swap** (i.e., **TokenExchange**) events for Curve **stETH-ETH** pool. We use an Ethereum Geth node on a Linux machine running Ubuntu 22.04 LTS, which is equipped with AMD 48-core CPU, 256 GB of RAM, and 12×2 TB SSD. We capture all the targeted events from block 11,473,216 (Dec 17, 2020) to block 17,866,191 (Aug 7, 2023), 963 days in total. We identify 290,984 **stake** events on Lido, 449,528 **deposit**, 238,388 **borrow**, 336,746 **withdraw** and 173,596 **repay** events on Aave V2 lending pool, and 105,310 **swap** events on Curve **stETH-ETH** pool.

Leverage Staking Detection. We proceed to analyze the users who adopt the direct or indirect leverage staking strategy. From the 449,528 **deposit** and 238,388 **borrow** events on Aave V2, we find that 743 addresses are used to deposit **stETH** as collateral and then borrow ETH. We then propose Algorithm 1 and 2 (see Appendix C) to identify the addresses involving

⁸<https://docs.aave.com/risk/liquidity-risk/borrow-interest-rate>

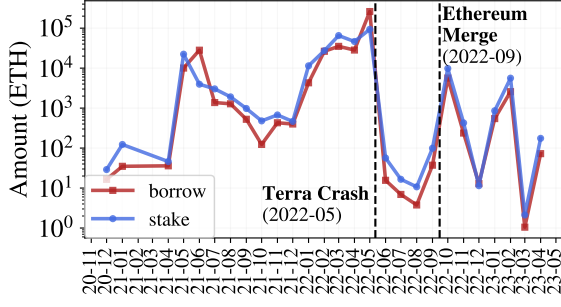


Figure 7: Direct leverage staking statistics.

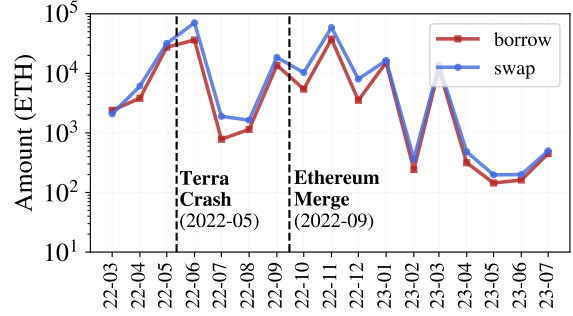


Figure 8: Indirect leverage staking statistics.

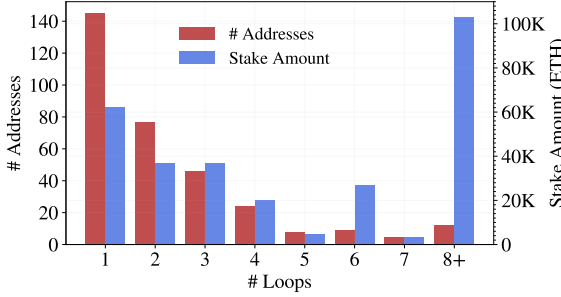


Figure 9: Stake amount and the number of direct leverage staking addresses by loops.

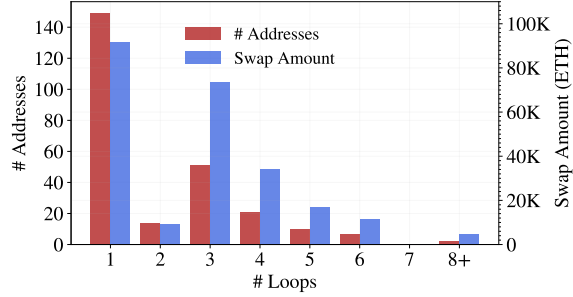


Figure 10: Swap amount and the number of indirect leverage staking addresses by loops.

direct and indirect leverage staking respectively. Specifically, we extract the event sequences of (stake, deposit, borrow, stake) and (swap, deposit, borrow, swap) in chronological order, which follows the direct and indirect leverage staking process (Figure 4).

We have identified 262 addresses that have been engaging in direct leverage staking activities, with a cumulative stake amount of 295,243 ETH. In addition, we observe 180 addresses that have performed the indirect leverage staking strategy, with a cumulative swap amount of 241,880 ETH. The distribution of leverage stake and swap amount is depicted in Figure 7 and 8 respectively. Interestingly, we observe that the volume of both direct and indirect leverage staking was substantially impacted by the Terra crash incident in May 2022. The stake amount of direct leverage staking experienced a drastic decline from the peak monthly stake amount of 93,661 ETH in May 2022 to 11 ETH in Aug 2022. Similarly, the swap amount of indirect leverage staking declines from 70,655 ETH in Jun 2022 to 1,639 ETH in Aug 2022. Later, the Ethereum Merge brought about a resurgence in leverage staking activities, with a stake amount of 9,814 ETH and a swap amount of 10,293 ETH in Nov 2022.

Leverage Staking Loops. Among 262 and 180 addresses that have adopted direct and indirect leverage staking, we conduct an analysis focusing on two key elements: the number of loops (denoted as n) and the leverage multiplier (denoted as $LevM_{(S,n)}$), derived from their extracted action sequence $\mathcal{E}_s$ (see Algorithm 1 and 2). To calculate the number of direct and indirect leverage staking loops, we identify consecutive sub-sequences in $\mathcal{E}_s$ consisting of (stake, deposit, borrow, stake) and (swap, deposit, borrow, swap) respectively. Figure 9 reveals that 145 addresses (55.35%) performed direct leverage staking with a single loop ($n = 1$), while Figure 10 shows that 149 addresses (82.78%) performed indirect leverage staking with a single loop. Although only a smaller subset of 12 addresses performs direct leverage staking with more than eight loops, their cumulative staking activities amount to a

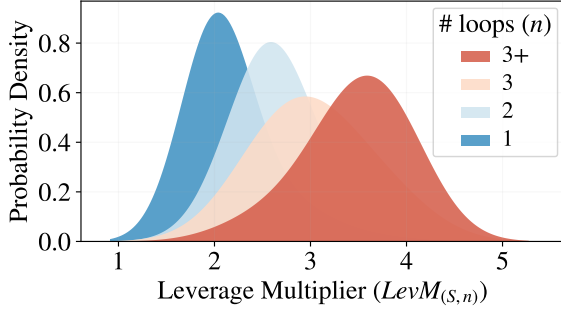


Figure 11: Distribution of direct leverage staking $LevM_{(S,n)}$ by loops.

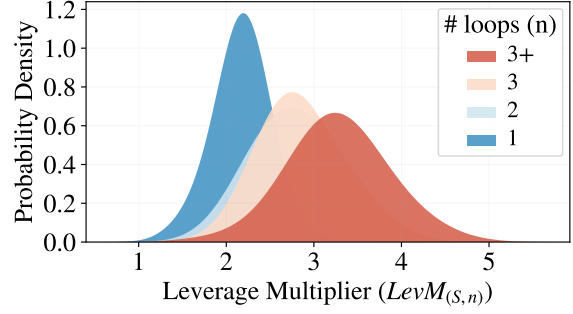


Figure 12: Distribution of indirect leverage staking $LevM_{(S,n)}$ by loops.

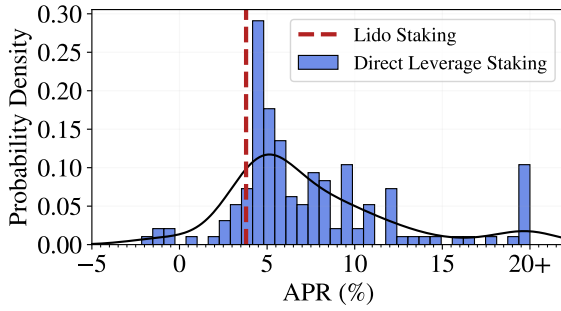


Figure 13: Direct leverage staking APR.

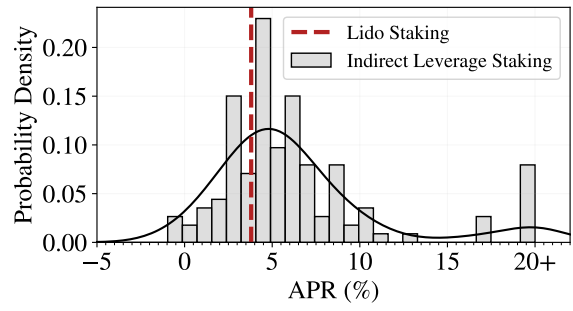


Figure 14: Indirect leverage staking APR.

significant volume of 102,998 ETH. This highlights a concentrated yet substantial engagement in direct leverage staking. In contrast, only 2 addresses performed indirect leverage staking with more than eight loops, with a total swap amount of 4,669 ETH. The difference in the number of participants and the total amount staked suggests distinct participant profiles and strategies between direct and indirect leverage staking. The comparison indicates that direct leverage staking is more likely to attract sophisticated market participants who are willing to perform more loops with substantial capital commitments.

Leverage Staking Multipliers. Furthermore, we compute $LevM_{(S,n)}$ for each address by taking into account the initial and the cumulative sum of **stake** or **swap** amount (See Equation 3). Figure 11 and 12 illustrate the distribution of $LevM_{(S,n)}$ across various n . The trend shows that a higher loop count n generally corresponds to higher $LevM_{(S,n)}$ in practical scenarios. Additionally, it is noteworthy that the majority (more than 90%) of the direct and indirect leverage staking addresses exhibit a $LevM_{(S,n)}$ smaller than 4.

Leverage Staking APR. We focus on a subset of 152 and 137 direct and indirect leverage staking addresses that have successfully repaid their debts and withdrawn their collateral from Aave **stETH-ETH** positions. To calculate their *actual* APR, as outlined in Equation 7, we consider the net earnings from **deposit** and **withdraw** actions, balanced against the ETH accrued through **borrow** and **repay** actions. Additionally, we account for the conversion of

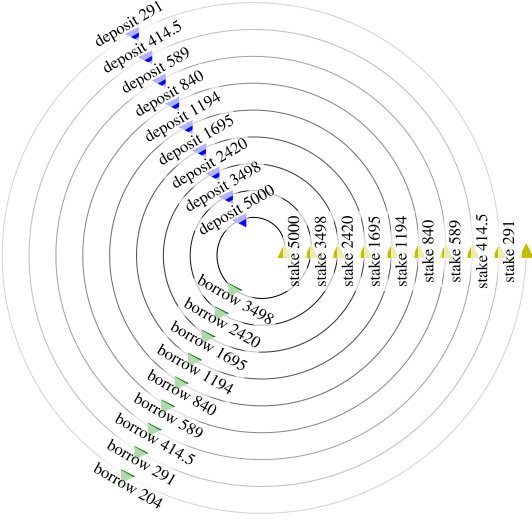


Figure 15: 0xD2...701: Direct leverage staking from block 14,617,906 to 14,627,202.

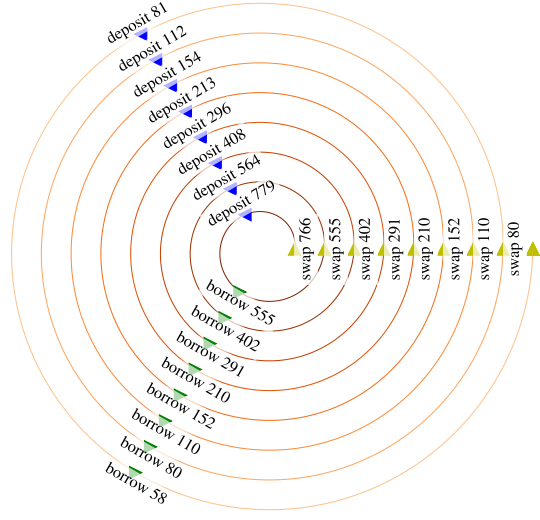


Figure 16: 0xA1...882: Indirect leverage staking from block 16,031,087 to 16,031,208.

accrued ETH to `stETH`, factoring in the `stETH` price at the time of the last `withdraw` action.

$$\text{actualAPR} = \frac{\left(\text{accruedstETH} - \frac{\text{accruedETH}}{p_{\text{stETH}}} \right) \cdot \frac{3600 \cdot 24 \cdot 365}{12}}{\text{totalDepositstETH} \cdot (\text{lastWithdrawBlock} - \text{firstDepositBlock})} \quad (7)$$

$$\text{accruedstETH} = \text{totalWithdrawstETH} - \text{totalDepositstETH}$$

$$\text{accruedETH} = \text{totalRepayETH} - \text{totalBorrowETH}$$

The distributions of direct and indirect leverage staking APR are visually depicted in Figure 13 and 14 respectively. Notably, our findings reveal that a significant majority (81.7%), precisely 137 (90.13%) direct and 99 (73.33%) indirect leverage staking addresses have realized an APR higher than the APR of conventional staking on Lido, highlighting the financial opportunities of such strategy.

Leverage Staking Examples. We present two examples to enhance the understanding of leverage staking. From block 14,617,906 to 14,627,202, a whale wallet address 0xD2...701 executed the direct leverage staking strategy with 9 loops. The recursive action sequences of (`stake`, `deposit`, `borrow`, `stake`) are shown in Figure 15. 0xD2...701 invested a principal amount of 5,000 ETH. The direct leverage staking results in a total investment amount of 16,145.5 ETH. This led to a leverage multiplier of 3.23, demonstrating the amplification effect of the leverage strategy. In another instance, from block 16,031,087 to 16,031,208, 0xA1...882 performed the indirect leverage staking strategy. This was accomplished by recursively executing the action sequence of (`swap`, `deposit`, `borrow`, `swap`). 0xA1...882 started with a principal investment amount of 766 ETH. Through 8 leverage loops, 0xA1...882 achieved a total investment of 2,624 ETH, resulting in a leverage multiplier of 3.43. This illustrates the effectiveness of the indirect leverage staking strategy in increasing the total investment.

7 Cascading Liquidation

In this section, we offer an overview of the `stETH` price deviation in the context of the Terra crash incident. We illustrate how the `stETH` price can potentially lead to liquidation cascades within the LSD ecosystem, particularly in the context of leverage staking.

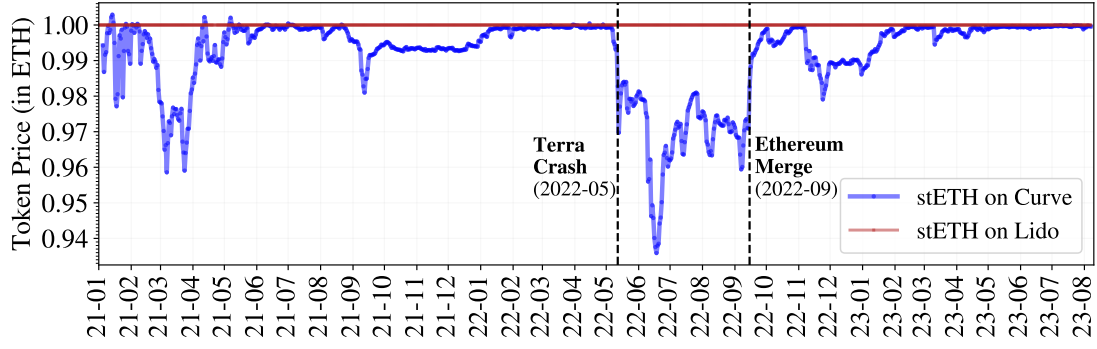


Figure 17: stETH price in the primary (Lido) and secondary (Curve) markets.

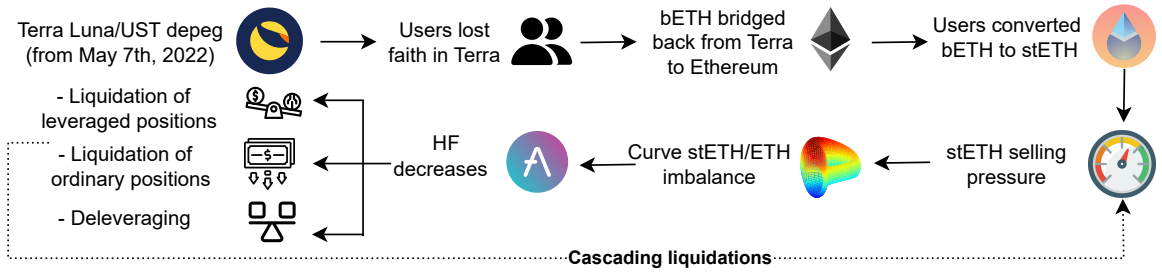


Figure 18: Illustration of liquidation cascades.

7.1 stETH Price Deviation and Terra Crash Incident

As a rebasing LSD, stETH changes its token supply to distribute rewards to stakers (see Section 3.4). As such, the stETH to ETH price in the primary market (i.e., Lido) is 1. While stETH is not required to trade on par with ETH in the secondary market (e.g., Curve), the price is anticipated to converge to 1. Our empirical data show that stETH did maintain a loose peg to ETH for most of its history. However, the stETH price began to drop from May 12, 2022, reaching its lowest point of 0.931 on May 18, 2022 (see Figure 17).

The stETH price decline can date back to the UST/LUNA depeg. The Terra collapse instilled fear and triggered selling pressure throughout the market [27, 28]. Specifically, following the UST/LUNA depeg incident between May 7 to 16, investors grew concerned about the security and stability of the Terra network. Given the prevailing bearish sentiment, investors moved to bridge back bETH (a wrapped version of stETH on Terra) from Terra to Ethereum via the Wormhole contract. Our data show that 614k bETH was bridged to Ethereum, with a remarkable 98% of these bETH converted back to stETH. This mass conversion reflects the widespread desire to exit Terra-based staking assets. Subsequently, the secondary market experienced significant selling pressure, primarily from institutions such as Celsius. This imbalance in the Curve stETH-ETH pool contributed to the further decline of stETH price.

7.2 Cascading Liquidation and User Behaviors

The decline in stETH price may trigger liquidation cascades within the LSD ecosystem, especially in the context of leverage staking (see Figure 18). Specifically, the decline in stETH price reduces the HFs of stETH collateralized borrowing positions on Aave, potentially leading to liquidations. In response to liquidations, users with leverage staking positions can either take no action or choose to deleverage their positions.

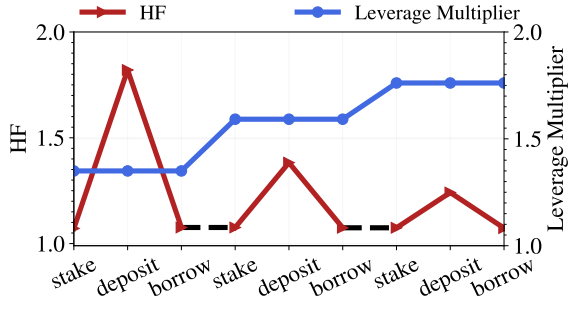


Figure 19: Example of the leverage action.

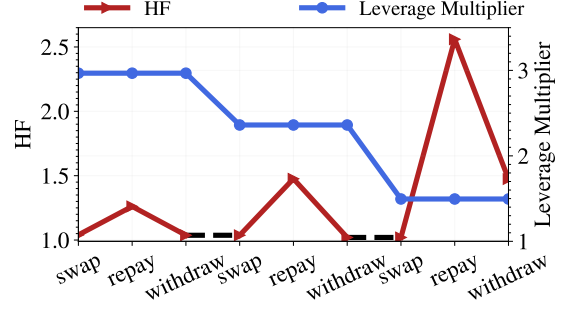


Figure 20: Example of the deleverage action.

On the one hand, users with leverage staking positions may take no action when their HF's approach the critical threshold of 1. In this case, their collateralized **stETH** might be liquidated. The liquidators repay **ETH** to acquire **stETH**, with the liquidation amount being amplified by $LevM_{(S,n)}$. Subsequently, a significant amount of **stETH** is sold in the Curve pool, contributing to additional selling pressure on **stETH** (see Figure 18). This extensive selling further imbalances the Curve **stETH**–**ETH** pool, resulting in a further decline in **stETH** price. Consequently, an increasing number of positions, including both leverage staking and ordinary positions, are vulnerable to liquidation as a result of declining HF's.

On the other hand, users can choose to deleverage their positions and restore HF's. Assuming U_i has executed a direct or indirect leverage staking strategy with n loops, U_i can initiate a deleveraging process with the following steps. (i) U_i executes a swap to convert **stETH** into **ETH** within the Curve **stETH**–**ETH** pool. (ii) The received **ETH** is then used to repay the **ETH** borrowed in the n^{th} loop. (iii) U_i then withdraws the **stETH** that was supplied in the n^{th} loop from Aave and continues converting it into **ETH** using the Curve pool. This “swap-repay-withdraw” process is repeated as necessary to restore HF's.

Taking address 0xD2...701 as an example, the overall trend of HF and $LevM_{(S,n)}$ during the deleveraging process (see Figure 20) exhibit a remarkable degree of symmetry when compared to those observed in the leveraging process (Figure 19). With each **repay** action, the HF of the address increases, as indicated by the red line, while the leverage multiplier decreases, as shown by the blue line. This symmetrical trend illustrates the correlation between repaying debt and improving HF, which consequently reduces $LevM_{(S,n)}$.

During the period from May 8, 2022 to May 18, 2022, i.e., the first ten days after the Terra crash (Figure 17), we observed 13 users actively deleveraging their direct leverage staking positions and 5 users deleveraging their indirect leverage staking positions. This activity resulted in a total debt repayment of 74,983.6 **ETH** and 61,085.5 **ETH** respectively. However, it is important to note that even if a user manages to avoid liquidation by deleveraging, the additional selling pressure generated by the **swap** transactions on Curve can still intensify the decline in **stETH** price. This decline may increase the vulnerability of other leverage staking and ordinary positions to liquidation. Such contagion effect may increase market instability and affect a broader range of participants beyond those directly engaged in deleveraging.

To summarize, users with leverage staking positions can take various actions to respond to potential liquidations. Regardless of their choices, these actions may contribute to additional selling pressure on **stETH**, further exacerbating price declines and liquidation cascades. This dynamic underscores the interconnection of leverage staking and the broader market ecosystem. In the following section, we will conduct stress tests to evaluate such risks.

8 Stress Testing

8.1 Motivation

By crawling the `liquidationcall` events on Aave V2 lending pool from Dec 17, 2020 to Aug 7, 2023, we identify 18 liquidations for the positions where users supplied `stETH` to borrow `ETH`, 7 liquidations for direct leverage staking positions, and 2 liquidations for indirect leverage staking positions. This relatively low number of liquidations can be attributed to the fact that `stETH` has historically only experienced a modest price decline, with a low of 0.931. However, drawing from the LUNA-UST incident, we recognize that a token may become entirely devalued. Should `stETH` undergo a devaluation similar to that of LUNA, it could trigger a surge in liquidations. Therefore, it is crucial to conduct stress tests to assess the risk of cascading liquidations under the worst-case scenario.

8.2 Simulation

Motivated by these concerns, we perform stress tests on the Lido-Aave-Curve LSD ecosystem under extreme conditions, simulating potential liquidation events, selling pressures, and subsequent liquidation cascades triggered by a significant `stETH` devaluation. Our objective is to address the following simulation questions (SQs):

SQ1 How are leverage staking positions affected by `stETH` devaluation?

SQ2 How does the liquidation of leverage staking positions affect the price of `stETH`?

SQ3 How do leverage staking positions affect ordinary positions during `stETH` devaluation?

SQ4 What are the effects of deleveraging actions on `stETH` price and market participants?

We first categorize Aave collateralized `stETH` borrowing positions into two groups: the leverage staking group (G_L) and the ordinary group (G_O). We then simulate four distinct scenarios to investigate the answers to the corresponding SQs.

8.2.1 SQ1

SQ1 aims to simulate the impact of a significant `stETH` devaluation on leverage staking positions. We focus on the fluctuations in HF and record the number of liquidated positions.

Simulation Setup. We initialize the Curve `stETH-ETH` pool by forking its state at block 17,500,000 (Jun 17, 2023), with reserve of 265,972 `ETH` and 266,966 `stETH`. Subsequently, we mimic the institutional selling pressure (e.g., Celsius; see Figure 18) after the Terra crash incident by simulating a sale of 170,000 `stETH` on Curve. This sizable transaction causes a decline in the `stETH` price, resulting in a new exchange rate of 100 `stETH` = 90.52 `ETH`, denoted as $p_0 = 0.9052$. In addition, we initialize G_L with 262 direct and 180 indirect leverage staking positions, each with an `address` that we have detected in Section 6. For each position, the values of `totalDebtETH`, `totalCollateralETH`, and HF are set to the corresponding values recorded in the transaction logs of that position’s most recent borrowing transaction. Furthermore, the `stETHPrice` for all positions is initialized as p_0 .

Simulation Process. The simulation process consists of a series of sequential rounds. In each round, the `stETHPrice` for all positions is updated as the current `stETH` price in the Curve `stETH-ETH` pool. Subsequently, the HF for each position is recalculated, using the updated `stETHPrice`. If, at any point, a position’s HF drops below the threshold of $HF = 1$, a

simulated liquidation event is triggered. In this scenario, a designated liquidator steps in to settle the debt by repaying it in ETH. In return, the liquidator receives the collateral in **stETH**. All received **stETH** is converted to ETH in the Curve **stETH**–ETH pool, as shown in Figure 18. This process continues until no more liquidatable positions remain.

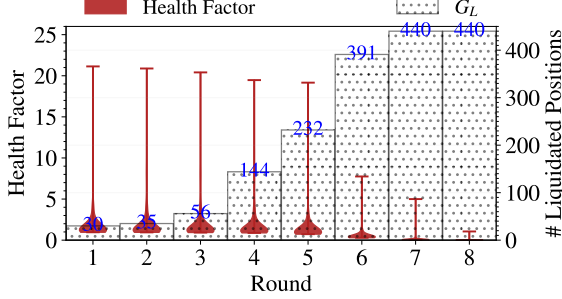


Figure 21: # liquidated leverage staking positions and the change of HFs.

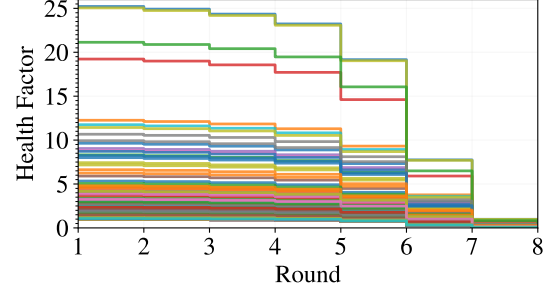


Figure 22: Simulated change in HFs for leverage staking positions in G_L .

Finding 1 During the devaluation of **stETH**, leverage staking positions become vulnerable to liquidations due to significant declines in their HFs.

Simulation Result. We examine the liquidation dynamics of G_L ’s leverage staking positions and the fluctuations in their HFs throughout simulations. Figure 21 shows the number of liquidated users in group G_L and the variations in HFs across different simulation rounds. The simulation terminates after 8 rounds, where 440 (99.55%) positions are liquidated. A noteworthy observation is that HFs of all positions exhibit a steep decrease, as depicted in Figure 22. The liquidation cascades experienced by leverage staking positions result in a total liquidated amount 497,375 ETH, ultimately driving the **stETH** price down to 0.01 ETH.

8.2.2 SQ2

SQ2 aims to explore how the liquidation of leverage staking positions can impact the price of **stETH**. We first simulate a scenario to evaluate the effects of leverage staking strategies (G_L), followed by a contrasting scenario where users do not adopt these strategies. The selling pressure originates from the liquidation of the corresponding positions.

We first simulate a scenario to assess the impact of leverage staking on **stETH** price and the liquidation volume. Next, we simulate an alternative scenario in which users within group G_L , which includes 262 direct and 180 indirect leverage staking positions, do not adopt leverage staking strategies. This involves setting the initial values for **totalCollateralETH**, **totalDebtETH**, and HF as the values recorded in the transaction logs when the *first* borrowing action for the position occurred.

Finding 2 Leverage staking amplifies the risks of cascading liquidations. The liquidation of leverage staking positions introduces additional selling pressure to the market, thereby exacerbating the decline in **stETH** prices and triggering further liquidations.

Figure 23 illustrates the comparative simulation results for scenarios where users adopt or do not adopt the leverage staking strategy. In the absence of leverage staking, the **stETH** price stabilized at 0.84 ETH in the last rounds, leading to a comparatively modest liquidation amount of 28,201 ETH. However, with the application of leverage staking, the **stETH** price plummeted to 0.01 ETH at the end of the simulation, and the liquidation amount (497,375 ETH)

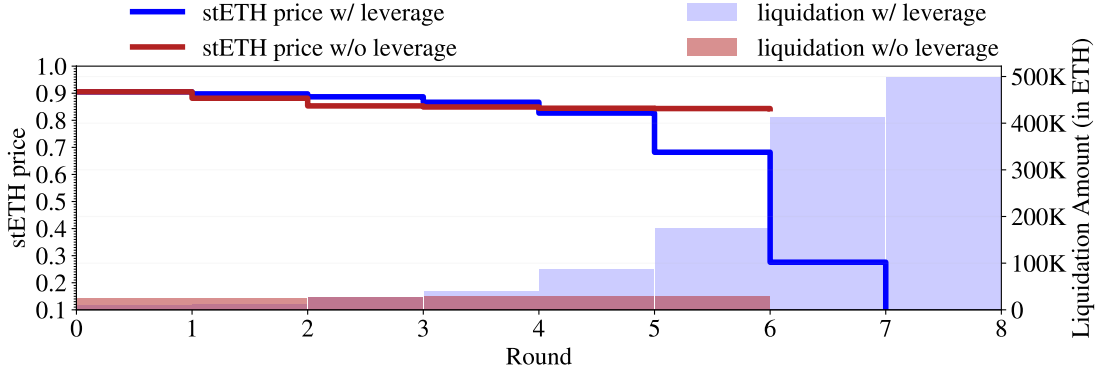


Figure 23: Comparison of stETH price and liquidation amount with and without leverage staking. The blue and red lines (bars) show the change of the stETH price (the change of liquidation amount) when users in G_L adopt or do not adopt the leverage staking strategy.

escalated to 16 times that of the scenario where such strategies were not applied. Our simulation findings indicate that adopting leverage staking strategies significantly exacerbates the risk of cascading liquidation in response to market downturns. This underscores the importance of robust risk management within the LSD system. Implementing such strategies without considering their potential impact on market stability can lead to adverse outcomes that affect a wide range of stakeholders.

8.2.3 SQ3

SQ3 aims to explore the impact of leverage staking positions on ordinary positions during significant stETH devaluation. This simulation constructs two scenarios: a control scenario, which involves only the ordinary group (G_O) consisting of 442 users, and an experimental scenario, where G_L (442 users) and G_O (442 users) coexist on the Aave platform. Both scenarios are subjected to identical simulation processes to record the number of liquidated positions within G_O and the fluctuations in the stETH price, allowing for a direct comparison of outcomes with and without the influence of leverage staking.

Finding 3 Leverage staking introduces broader systemic risks because it significantly exacerbates the liquidation of ordinary positions during stETH devaluation.

The simulation results for the control scenario (see Figure 24) indicate that 75 ordinary positions are liquidated in the absence of G_L . In contrast, the results for the experimental scenario (see Figure 25) reveal that 440 ordinary positions are liquidated, suggesting a significant increase in liquidations when G_L is present. Our simulation results suggest that leverage staking not only intensifies the risk profile of individual portfolios but also contributes to broader systemic risks, particularly during periods of sharp declines in LSD prices. The comparative analysis of liquidation rates between the two scenarios underscores the influence that leverage staking positions can exert on the stability of ordinary positions.

The increased liquidations in the presence of G_L imply a contagion effect, where vulnerabilities in leveraged positions can cascade to affect even traditionally less risky, ordinary positions. This suggests that systems designed to stabilize market dynamics need to account not only for individual positions but also for their interdependencies. Therefore, our simulation results highlight the necessity for regulatory frameworks and platform governance structures to consider these interconnections.

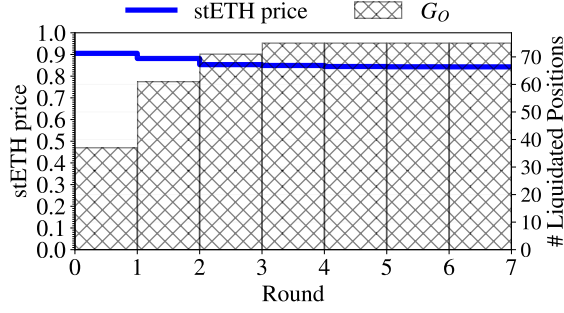


Figure 24: The change of the stETH price and # liquidated positions in G_O in the absence of the leverage staking group (G_L).

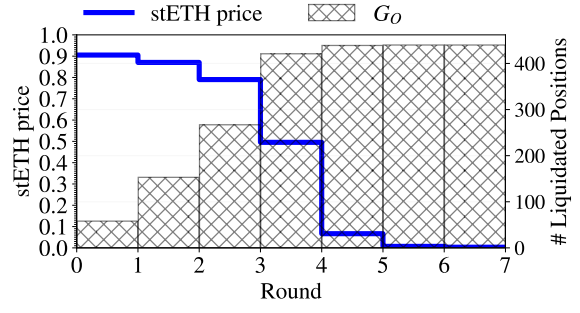


Figure 25: The change of stETH price and # liquidated positions in G_O in the presence of the leverage staking group (G_L).

8.2.4 SQ4

As discussed in Section 7.2, users holding leverage staking positions might choose to deleverage during a decline in stETH value. SQ4 is designed to examine the effects of such deleveraging actions on the stETH price and LSD market participants. We simulate two scenarios: the control scenarios where G_L does not deleverage and the experimental scenarios where G_L chooses to deleverage at the beginning of the simulation (round 0).

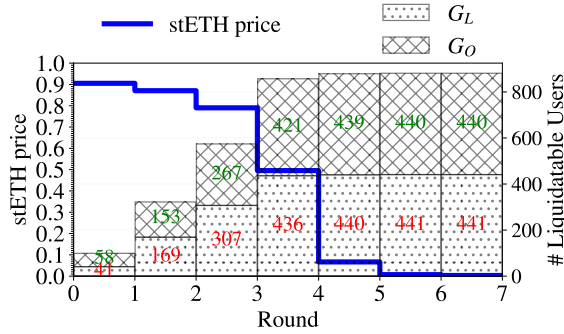


Figure 26: The change of stETH price and # liquidated positions without deleveraging.

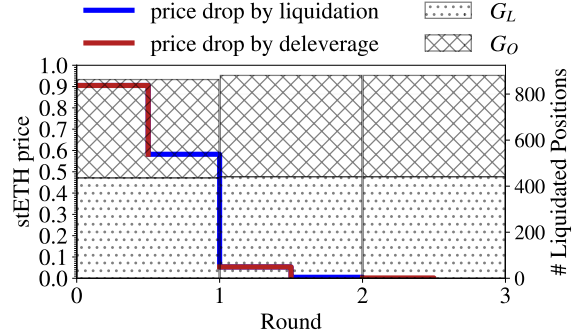


Figure 27: The change of stETH price and # liquidated positions with deleveraging.

Finding 4 The deleveraging actions taken by G_L can introduce additional selling pressure, thereby intensifying the liquidation cascades among system participants.

When users in G_L do not choose to deleverage, our simulation result (see Figure 26) shows that the liquidation ends in 7 round, with 441 users in G_L and 440 users in G_O being liquidated. Conversely, when G_L decides to deleverage at round 0, the liquidation process is significantly shortened, ending in just 3 rounds. This pronounced difference underscores the critical role that deleveraging actions play in market dynamics. Not only do they shorten the duration of liquidations, but they also potentially amplify market volatility by introducing additional selling pressure. This indicates that deleveraging action can exacerbate systemic risk by accelerating the liquidation cascade, affecting a broader range of system participants.

9 Discussion and Future Research Directions

Our comprehensive stress tests on the Lido-Aave-Curve LSD ecosystem reveal critical vulnerabilities and dynamic interplays under extreme conditions of significant `stETH` devaluation. These simulations illustrate that leverage staking strategies, while innovative, expose the market to heightened risks. The presence of leverage staking significantly escalates the risk of cascading liquidations within the LSD ecosystem. This finding underlines a crucial concern: systemic risk is exacerbated not only through direct liquidations but also via the market pressures these actions generate. The selling pressure on `stETH`, driven by both liquidations and deleveraging actions, can trigger a contagion effect across the system, further depressing `stETH` prices and adversely impacting the financial stability of broader market participants, including ordinary users. Therefore, it is crucial to strike a balance between leveraging opportunities for higher returns and the potential for destabilization in LSD ecosystems.

Building upon these insights, future research can pursue several avenues. A crucial direction is the development of refined models that simulate a broader range of conditions, incorporating more granular behaviors of market participants and liquidity variations. This could lead to more robust parameterization of platforms such as Aave, similar to the “safe parameterization” design used in traditional finance, which aims to mitigate risks without stifling innovation. Additionally, exploring new regulatory frameworks tailored to LSDs could help mitigate the systemic shocks observed in our simulations. By integrating advanced risk management strategies and regulatory innovations, future research can contribute to creating a more resilient LSD ecosystem. This involves a holistic approach to understanding the interdependencies and collective behaviors that drive the resilience of these platforms.

10 Conclusion

This work systematically examines the leverage staking strategy with LSDs, providing both analytical and empirical insights into its opportunities and risks. Our analytical study introduces a formal model that captures both direct and indirect leverage staking strategies, establishing key metrics such as leverage amounts, loops, multipliers, and APRs. Our empirical study builds heuristics to identify historical leverage staking positions and evaluate their performance. Notably, we observe that the majority of these positions yield an APR exceeding that of conventional staking, highlighting their high-return potential.

However, these financial opportunities come with inherent risks. To address this, we conduct stress tests under extreme market scenarios to assess the vulnerabilities of leverage staking. Our findings reveal that leverage staking significantly amplifies the risk of cascading liquidations, leading to intensified selling pressure and systemic instability. Additionally, it propagates a contagion effect, exacerbating the liquidation of ordinary positions and posing broader systemic risks to the LSD ecosystem.

We hope this work serves as a foundation for academic researchers and protocol developers to design robust risk assessment frameworks and implement safe parameterizations, ultimately enhancing the resilience and sustainability of the LSD ecosystem.

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A Aave Parameter Configuration

Table 3 depicts the historical changes of Aave parameter configurations. We crawl the `collateralConfigurationChanged` events for Aave V2 lending pool.

Block Number	Transaction Hash	LTV(l)	LT
14289297	0x94780dc1914af5aec3b6d303e2d974669074bbceb6d1baac7d93ad0400593db0	0.70	0.75
14693506	0x993926559f17a579c3b0c0b5fc83fd3c9d5772e9b314f7f0e78e704c6b984726	0.73	0.75
14804760	0xbc40546b65ada9f5d4f8346f405a5f9c0da6d8f66bb27b7c64c0efa70eeae080	0.69	0.81
14837221	0x5206c144845ac63f982125f51c31b0fc474655281e2d433f8270505d87f8bbf4	0.70	0.75
14999895	0x25e33e04d2e5d92acd91f542a2677045e59b2d6b385e4fa0315a404396bc5c99	0.69	0.81
15759644	0x48653014a79433bf5f21781424b306a697f4476077a8b92e17b2ae6eda58706e	0.72	0.83
16392718	0xdcaa33ddef0700a2a625e0b7a0c2da14499901e42c073064390f11dd83cefd19	0.69	0.81

Table 3: Historical changes of Aave V2 parameter configurations.

B Generalized Formalization For Leverage Staking

B.1 Generalized Formalization

In addition to the standardized cases discussed in Section 5, real-world leverage lending situations can exhibit substantial variation among users. Specifically, we delineate the following variations using the direct leverage staking strategy as an example.

(i) Within each leverage staking loop, U_i may choose not to supply all the `stETH` acquired from Lido as collateral on Aave. Instead, in the k^{th} loop, U_i may opt to supply only c_k ($c_k \in [0, 1]$) percent of the `stETH` amount. (ii) In the k^{th} loop, U_i has the option to borrow an amount of `ETH` that is less than the maximum borrowing capacity. In this scenario, U_i 's effective borrowing capacity becomes $b_k \cdot l$, where $b_k \in [0, 1]$. (iii) In the k^{th} loop, U_i has the flexibility to restake part of the borrowed `ETH` on Lido. U_i may choose to restake only s_k ($s_k \in [0, 1]$) percent of the `ETH` borrowed in the $(k - 1)^{\text{th}}$ loop (or the principal amount when $k = 1$) at the start of the k^{th} loop. After borrowing `stETH` from Aave, U_i may restake only s_{k+1} ($s_{k+1} \in [0, 1]$) percent of the borrowed `ETH` at the end of the k^{th} loop (equals to the beginning of $(k + 1)^{\text{th}}$ loop). Note that as illustrated by Figure 4, the stake and restake parameters (s_k and s_{k+1}) together establish the k^{th} loop. Equation 8 introduces a generalized formalization to accommodate these variations.

$$\begin{aligned}
A_{(S,n)} &= S \cdot \sum_{k=1}^{n+1} \left(\prod_{i=1}^k s_i \right) \cdot \left(\prod_{i=1}^{i-1} c_i \right) \cdot \left(\prod_{i=1}^{k-1} b_i \right) \cdot \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^{k-1} \\
C_{(S,n)} &= S \cdot \sum_{k=1}^n \left(\prod_{i=1}^k s_i \right) \cdot \left(\prod_{i=1}^k c_i \right) \cdot \left(\prod_{i=1}^{k-1} b_i \right) \cdot \frac{(l \cdot p_{t_0}^a)^{k-1}}{(p_{t_0}^m)^k} \\
B_{(S,n)} &= S \cdot \sum_{k=1}^n \left(\prod_{i=1}^k s_i \right) \cdot \left(\prod_{i=1}^k c_i \right) \cdot \left(\prod_{i=1}^k b_i \right) \cdot \left(\frac{l \cdot p_{t_0}^a}{p_{t_0}^m} \right)^k \\
HF_{U_i}(p_{t_c}^a | p_{t_0}^a) &= \frac{C_{(S,n)} \cdot p_{t_c}^a \cdot \text{LT}}{B_{(S,n)}}
\end{aligned} \tag{8}$$

C Leverage Staking Detection Algorithm

Algorithms 1 and 2 depict the heuristics used to detect addresses that have performed direct and indirect leverage staking respectively.

Algorithm 1: Direct Leverage Staking Detection.

Input: An address `addr`
Output: `addr`'s leverage staking actions.

- 1 Extract `addr`'s **deposit** events $\{w_i\}_i$ and **borrow** events $\{b_j\}_j$ on Aave, and **stake** events $\{s_k\}_k$ on Lido;
- 2 Let $\mathcal{E} = \{w_i\}_i \cup \{b_j\}_j \cup \{s_k\}_k$;
- 3 Convert $\mathcal{E}$ to a sequence $\mathcal{E}_s$ by sorting $\mathcal{E}$ in chronological order;
- 4 **if** $\mathcal{E}_s$ contains a sub-sequence with a order of (*stake, deposit, borrow, stake*) **then**
- 5 **if** For the sub-sequence (*stake₀, deposit, borrow, stake₁*): (i) The *stETH* amount received in *stake₀* event $\approx$ the *stETH* amount in *deposit* event; (ii) The *stETH* amount in *deposit* event $>$ the *ETH* amount in *borrow* event; (iii) The *ETH* amount in *borrow* event $\approx$ the *ETH* amount in *stake₁* event **then**
- 6 **return** $\mathcal{E}_s$;
- 7 **return** $\emptyset$;
- 8 **else**
- 9 **return** $\emptyset$;

Algorithm 2: Indirect Leverage Staking Detection.

Input: An address `addr`
Output: `addr`'s indirect leverage staking actions.

- 1 Extract `addr`'s **deposit** events $\{w_i\}_i$ and **borrow** events $\{b_j\}_j$ on Aave, and **swap** events $\{s_k\}_k$ on Curve;
- 2 Let $\mathcal{E} = \{w_i\}_i \cup \{b_j\}_j \cup \{s_k\}_k$;
- 3 Convert $\mathcal{E}$ to a sequence $\mathcal{E}_s$ by sorting $\mathcal{E}$ in chronological order;
- 4 **if** $\mathcal{E}_s$ contains a sub-sequence with a order of (*swap, deposit, borrow, swap*) **then**
- 5 **if** For the sub-sequence (*swap₀, deposit, borrow, swap₁*): (i) The *stETH* amount received in *swap₀* event $\approx$ the *stETH* amount in *deposit* event; (ii) The *stETH* amount in *deposit* event $>$ the *ETH* amount in *borrow* event; (iii) The *ETH* amount in *borrow* event $\approx$ the *ETH* amount in *swap₁* event **then**
- 6 **return** $\mathcal{E}_s$;
- 7 **return** $\emptyset$;
- 8 **else**
- 9 **return** $\emptyset$;
